# Shear-Layer Instabilities: Particle Image Velocimetry Measurements and Implications for Acoustics

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## Abstract

The use of particle image velocimetry (PIV) to study the spatial and temporal features of unsteady fluid flow has increased dramatically in the past five to ten years. One particular application of PIV is to examine how shear-layer instabilities and turbulence lead to radiated sound. In this review, the basic operation of a PIV system is provided along with an introduction to the equations that relate unsteady fluid motion to sound. The references then illustrate how PIV is currently used in a number of canonical flow problems of interest in which the phenomena are dominated by shear-layer instabilities that lead to radiated noise. Specifically, cavity flows, flow over airfoils and cylinders, and finally jet flows are considered.

## 1. INTRODUCTION

Sound generated by fluid flow, or aeroacoustics, is an important topic in the subject of fluid dynamics. Sound is often an undesired by-product of fluid flow in engineered devices. Examples include fan noise and jet noise from an aircraft. The mechanisms that govern how sound is generated by flow have been studied intensively since the original work of Lighthill (1952). This landmark paper showed that sound is produced whenever unsteady fluid motions exist. Although there have been significant advancements in the basic principles of sound generation, the complicated and poorly understood characteristics of most fluid flows have made it difficult to understand and predict the sources of sound from even simple flows of interest.

Particle image velocimetry (PIV) is an experimental method in which a fluid region is illuminated with a pulsed laser and imaged. Tracer particles are used to follow the fluid velocity and reflect the laser light into the camera. These images are processed to provide an instantaneous map of the velocity field. Significant advances have been made in PIV hardware and software in recent years. The resulting accuracy, spatial resolution, commercial availability, and ease of use of PIV systems have led to a dramatic increase in the use of PIV for the study of many fluid flows.

In recent years, PIV has been utilized in the study of the unsteady fluid motions that are linked to sound production. Although the specific uses of PIV data vary considerably, the general principle is to obtain spatially resolved (and often time-resolved) velocity field information and apply the results in the context of the equations that relate the velocity field to the far-field sound pressure. Understanding how this type of analysis is typically accomplished will require a basic understanding of PIV acquisition and data processing. This is presented in the following subsection. A review of the aeroacoustic equations that are often used in conjunction with PIV measurements is then presented. These two subsections set the context for understanding how flow instabilities and turbulence can lead to the production of sound and how PIV has been used in the study of sound sources. A third subsection describes a relatively new use of PIV data for estimating fluid pressure. This method has the potential to improve our understanding of sound production in many flows in which unsteady surface pressure is of primary importance.

The remaining sections of this review describe recent results and discoveries from three flow types: cavities, external aerodynamic flows, and jet noise. These flows were chosen because they each exhibit a specific mechanism in which shear-layer instabilities and small-scale turbulence lead to sound production. These flows also have relatively simple geometries and thus serve as archetype configurations for how PIV measurements can impact our understanding of how coherent motions and turbulence can create sound.

### 1.1. Particle Image Velocimetry Measurement Techniques

The PIV measurement technique has been used in its current form for nearly 30 years and is the subject of a number of books and review articles (see, e.g., Adrian 1991, Adrian 2005, Raffel et al. 1998). Only a simple description of PIV is provided here for completeness in understanding how current PIV measurements are used for the study of unsteady flows and acoustics.

A schematic of the basic hardware setup of typical, planar PIV is shown in **Figure 1**. The flow is seeded with small particles used to reflect the laser light. The diameter of the particles varies depending on the application, but generally is of the order of 1–3 μm. Smaller particles provide better tracking of the true fluid velocity (through Stokes drag) but will provide lower light intensity to the camera. The particles are typically illuminated with a pulsed laser system. Planar PIV generally requires a cylindrical lens to form a light sheet that can be imaged with one or more cameras. Two laser pulses are used to illuminate the particles. The images are recorded

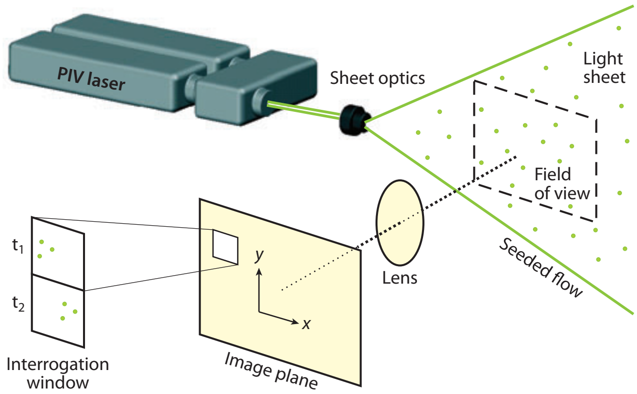



**Figure 1**

Schematic of a planar particle image velocimetry (PIV) system. Figure courtesy of LaVision, Inc.

with either film-based or digital cameras and stored. Postprocessing is then used to divide the field of view into square interrogation windows. The displacement of the particles is then estimated using various types of image correlation algorithms. The displacement is then divided by the time elapsed between the two laser pulses to arrive at a single velocity vector for each interrogation window.

The results provided by the standard, planar PIV system are realizations of two components of the velocity vector in the plane of the laser sheet [e.g., $u(x,y)$, $v(x,y)$, where the $x$-$y$ plane is defined by the sheet of the laser]. Most often these measurements are obtained with a dual-cavity pulsed Nd-Yag laser. These systems have the advantage of high power in a small, compact system. However, the maximum pulse rate is typically approximately 15 Hz, which results in measurements that are statistically independent in time for most flows of interest.

A large number of advances have been made that expand the capabilities of PIV beyond the standard, planar measurements depicted in **Figure 1**. Stereo-PIV requires two cameras that are focused at an oblique angle to the laser sheet. This method provides three components of velocity to be acquired over the planar domain defined by the laser sheet and the intersection of the field of view of the two cameras. The measurement accuracy of stereo-PIV is generally lower than two-component PIV. Data transfer rates from the two cameras to the computer can also limit the effective sampling frequency to less than the 15 Hz allowed by the laser system.

The fluid velocities in most flows that radiate significant sound levels are in the range of tens to hundreds of meters per second. Given the low-acquisition frequency of PIV systems, it is not possible to obtain time-resolved or frequency-domain information about the flow. Data analysis often either uses information derived from the instantaneous vector maps or computes phase-averaged quantities based on a large-scale instability or periodic signal.

Time-resolved PIV systems have recently become commercially available that allow planar measurements to be obtained at frequencies of 10 kHz or more. However, there are a number of current limitations to these systems. For example, the high-frequency laser systems are relatively low power and thus limit the size of the camera's field of view. Additionally, the high-speed cameras generally have lower light sensitivity, which makes high-quality PIV measurements more difficult. It is imagined that hardware and software advances will eventually resolve many of these issues, and time-resolved PIV will provide a significant tool for the study of acoustic sources.

Tomographic PIV is a recently developed technique that uses three or four cameras with volume illumination to obtain all three components of the velocity vector in a volume of fluid. Although the data-acquisition rates for tomographic PIV are generally even lower than standard PIV, the advantage is that the three-component measurements in a volume allow for all the acoustic source terms to be computed as well as their two-point correlation. The importance of this is discussed in the following subsection. An edited review of many of the latest advances in PIV measurement techniques can be found in Schroder & Willert (2008).

## 1.2. Flow-Induced Sound Theory

The sections that follow focus primarily on recent research in which flow instabilities and turbulence lead to the direct radiation of sound. Sound generated from flow-induced vibrations is also an important area of research but is not considered explicitly here. The equations that govern flow-induced noise from rigid surfaces can be found in a number of references, notably Goldstein (1976) and Blake (1986).

Lighthill originally rewrote the momentum equation as

$$\frac{\partial^2 \rho}{\partial t^2} - c_0^2 \nabla^2 \rho = \frac{\partial^2 T_{ij}}{\partial y_i \partial y_j}, \tag{1}$$

where

$$T_{ij} = \rho u_i u_j + (p - c_0^2 \rho)\delta_{ij} + \tau'_{ij} \tag{2}$$

is the Lighthill stress tensor, where $c_0$ represents the speed of sound in the fluid. The tensor $u_i u_j$ is the Reynolds stress. The term $\tau'_{ij}$ is the viscous part of the Stokes stress tensor, and the term $p - c_0^2 \rho$ represents the difference between the pressure fluctuations and those that would be caused by an isentropic density fluctuation. The pressure $p$ and density $\rho$ represent the local instantaneous pressure and density of the fluid, and thus contain the acoustic fluctuations in the far field. Most often the Reynolds stresses are the largest component of the Lighthill stress tensor, and Equation 1 may be simplified to

$$\frac{\partial^2 \rho}{\partial t^2} - c_0^2 \nabla^2 \rho = \frac{\partial^2 (\rho u_i u_j)}{\partial y_i \partial y_j}, \tag{3}$$

where the right-hand side represents the nonlinear fluid motions in a turbulent flow. Equation 3 reduces to the wave equation of linear acoustics theory at any location where $u_i u_j = 0$. Thus knowledge of the spatial and temporal characteristics of $u_i u_j$ is crucial for the modeling of acoustic radiation. In open flows, such as jets, Equation 3 is often used to describe the radiated sound by considering the right-hand side to be defined as an acoustic source that is nonzero in the near-field region in the jet. Outside the jet the fluid is assumed to be at rest and satisfies the homogeneous wave equation for acoustic pressure.

In situations in which surfaces are present in a flow, Curle (1955) provided a derivation of Lighthill's equation to account for the effects of surface pressure. Again the reader is referred to Goldstein (1976), Blake (1986), and the original papers of Curle (1955) and Powell (1960) for more details. A useful form of the result is given as

$$\nabla^2 \rho - \frac{1}{c_0^2}\frac{\partial^2 \rho}{\partial t^2} = -\frac{\partial q}{\partial t} + \frac{\partial F_i}{\partial y_i} - \frac{\partial^2 T_{ij}}{\partial y_i \partial y_j}, \tag{4}$$

where $q$ is the rate of mass injection per unit volume, and $F_i$ is the $i$-th component of the force vector per unit volume. In this form, the three terms on the right-hand side of Equation 4 appear as idealized monopole, dipole, and quadrupole sources. This formulation is particularly important for the interpretation and application of PIV data for understanding acoustic sources. For example, flow over a Helmholtz resonator, as reviewed below, can be considered to be an unsteady injection of mass at the orifice when the size of the orifice is small compared to the acoustic wavelength. Similarly, in external flows, such as an airfoil or cylinder, the net unsteady force created on the object by the unsteady pressure field can be considered as the acoustic source. It is generally true that the contribution of the stress tensor term to the net radiated sound will be small compared to the unsteady forces when the Mach number is small (e.g., $M < 0.5$). At higher speeds, both the stress tensor and the aerodynamic forces can be important.

Powell (1960) and later Howe (1998) provided another simplified form in which the unsteady distribution of vorticity was considered as the sound source. Howe's equation for isentropic, low–Mach number flows is given as

$$\frac{1}{c_0^2}\frac{\partial^2 B}{\partial t^2} - \nabla^2 B = \nabla \cdot (\omega \times u)\,, \tag{5}$$

where $B = p/\rho + u^2/2$ is the acoustic variable representing the total enthalpy of the flow, and $\omega$ is the vorticity vector. Equation 5 is of particular interest as regions of intense vorticity are often spatially concentrated and can be measured in detail with PIV.

It is worth noting that all the various forms of the governing equations are similar, in that the left-hand side appears as a linear wave operator, and the right-hand side represents the non-linear fluid motions that are interpreted as a source of aerodynamically generated sound. With this interpretation, there are several common solution methods to all these forms. One general methodology is through the use of a frequency-domain Green's function, $G(\boldsymbol{x}, \omega)$. Specifically, let us consider a generalized source term $\sigma(\boldsymbol{x}, \omega)$, defined as the Fourier transform of any one of the right-hand-side terms in Equations 1, 3, 4, and 5. The autospectral density of the far-field acoustic pressure can be given as

$$\Phi_{pp}(\mathbf{x}, \omega) = \iiint_V \cdot \iiint_V \Phi_{\sigma\sigma}(\mathbf{y}_1\mathbf{y}_2, \omega) G^*(\mathbf{x}, \mathbf{y}_1, \omega) G(\mathbf{x}, \mathbf{y}_2, \omega) d^3\mathbf{y}_1 d^3\mathbf{y}_2, \tag{6}$$

where $\Phi_{\sigma\sigma}(\mathbf{y}_1, \mathbf{y}_2, \omega)$ represents the two-point correlation function of the frequency-dependent source term. The Green's function can often be estimated analytically for simple geometries. In more complicated geometries, it can often be calculated or found empirically.

The radiated sound can be calculated using the Green's function solution given sufficient information regarding the unsteady velocity and pressure in the moving fluid. It is important to recognize, however, that a complete measurement of $\Phi_{\sigma\sigma}(\mathbf{y}_1, \mathbf{y}_2, \omega)$ in any turbulent flow is well beyond current experimental capabilities. Much of the research described in this review is focused on using PIV to obtain limited quantitative or qualitative information about $\Phi_{\sigma\sigma}(\mathbf{y}_1, \mathbf{y}_2, \omega)$ or the unsteady flow from which it is calculated. For example, the standard planar velocity measurements described in Section 1.1 provide two components of velocity as well as their gradients in the plane of the measurement at a maximum acquisition frequency of 15 Hz. This type of information generally provides only a qualitative representation of where the acoustic sources are located. In flows that have a periodic or nearly periodic instability, phase averaging of the PIV results can provide more quantitative information about the acoustic sources at the instability frequency. Note also that the spatial scales of turbulence are generally available through two-point correlations.

## 1.3. Unsteady Surface Pressure from Particle Image Velocimetry

Unsteady surface pressure plays a particularly important role when the Mach number is moderate or low. In this case, the direct radiation from the quadrupole term is small compared with that from the surface pressure term. If the object under consideration can be considered to be compact (i.e., the largest physical dimension of the object is small compared with the wavelength of sound), then the sound can be approximated as a simple dipole such that only the net unsteady force needs to be computed to estimate the radiated sound (i.e., the second term on the right-hand side of Equation 4). Thus, considering only the unsteady surface pressure around an object represents a significant simplification to the study of acoustic sources. Although PIV provides only velocity field information, a number of recent investigations have explored how PIV measurements can provide an indirect estimate of the instantaneous pressure field in a fluid. The advantage of PIV as compared to traditional pressure measurements (e.g., surface-mounted microphones) is that PIV provides an unobtrusive measurement of pressure that is spatially well resolved over a significant spatial domain.

Obtaining aerodynamic forces from PIV has been discussed by Noca et al. (1997), Lin & Rockwell (1996), and Unal et al. (1997). Unal et al.'s approach was to use a control volume along with the velocity vectors and their spatial derivatives to estimate the fluid forces. Note that the net aerodynamic forces are recovered using this method. Thus a dipole-type source can be resolved, but the details of the unsteady pressure distribution are not.

Fujisawa et al. (2005) employed PIV to calculate pressure from Poisson's equation using flow over a circular cylinder as a baseline flow for comparison. Surface pressures can also be estimated by providing a spatial representation of the instantaneous acceleration field and solving for the pressure using the momentum equation. Liu & Katz (2006) described a method using a two-camera, four-laser PIV technique that allowed the measurement of the acceleration vector in the plane of the measurement. The pressure gradient was then estimated by the acceleration and integrated spatially to obtain an estimate for a given point of interest. The path of integration was varied, and a mean value from the varied paths was used as the final estimate of the instantaneous pressure. Further refinements and an application of this method have been described by Jaw et al. (2009).

Van Oudheusden et al. (2006, 2007; van Oudheusden 2008) and Ragni et al. (2009) have developed processing methods to obtain surface pressure and net unsteady forces from PIV in both slow- and high-speed flows. They focused on the use of a NACA 0012 airfoil at Mach numbers up to 0.8 and angles of attack up to 8°. The lift, drag, and pressure obtained from the PIV results were compared with traditional surface pressure measurements.

The methods described all have significant potential for use in measuring aeroacoustic sources. Although differences between the various techniques are noted, they generally provide similar results, and the choice of one method over another seems to be motivated by the specifics of the geometry involved and the availability of PIV hardware [e.g., the acceleration measurements of Liu & Katz (2006) requires a two-camera, four-laser system].

## 2. CAVITY FLOW

The flow over an open cavity forms one of the canonical flows of interest in fluid mechanics. General reviews of research in these types of flows have been provided by Rockwell & Naudascher (1978, 1979) and more recently by Rowley & Williams (2006). The problem is of general interest because a resonance can occur in which the free shear-layer instability interacts with a feedback mechanism. There are three classifications of cavity flows that have been studied based on the type of feedback that occurs. The first is the study of flow in a shallow cavity with fully incompressible flow, in which the feedback mechanism is hydrodynamic in nature (that is, the pressure fluctuations travel at an effectively infinite speed). The second classification is the flow of a compressible fluid over a shallow cavity. Known as Rossiter's flow, the feedback mechanism is an acoustic disturbance that travels from the downstream edge of the cavity to the upstream separation point. The third classification is grazing flow over a Helmholtz resonator or deep cavity. Here the Mach number is again generally low, and the orifice region of the flow can be considered incompressible. However,

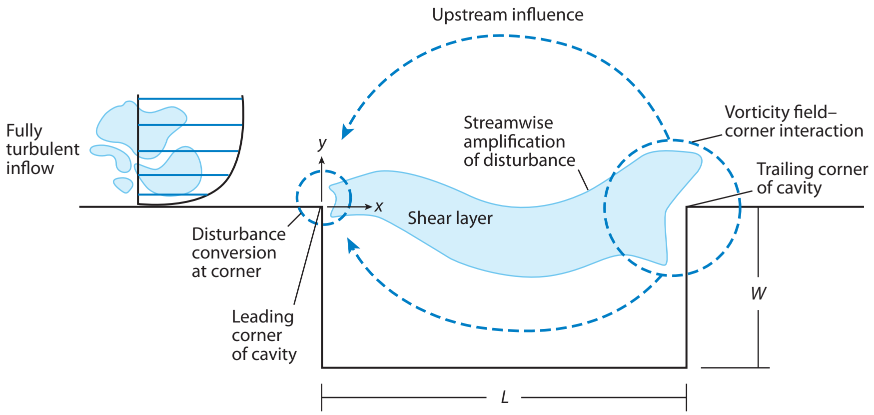



**Figure 2**

Schematic of the general shallow-cavity problem. Figure taken from Rockwell et al. (2003).

the presence of a large volume of compressible fluid will lead to a resonant frequency that permits large pressure fluctuations inside the resonator. Each of these flow types is described separately in the following subsections.

## 2.1. Shallow, Incompressible Cavity Flows

There exists a significant body of literature that is focused on the flow of an incompressible fluid over a shallow cavity. A schematic of the flow as given by Rockwell et al. (2003) is shown in **Figure 2**. In this flow, the approach boundary layer separates at the upstream sharp edge of the cavity. If the flow is strictly two dimensional, then the reattachment point is also fixed to the downstream edge of the cavity to ensure the flow inside the cavity is incompressible.

The instability of the free shear layer over the cavity often leads to periodic or nearly periodic oscillations in the velocity and pressure fields. The unsteady surface pressure can be considered as a boundary condition to the acoustic problem in which the cavity acts as a dipole-like source (Burroughs & Stinebring 1994). The frequency and amplitude of the resulting acoustic tones have been studied for many decades (see, e.g., Blake 1986, Zoccola & Farabee 2001).

PIV has been used extensively to investigate the mechanisms related to the hydrodynamic instability and resulting acoustic tones of shallow cavities. Much of the progress has been made by Rockwell and coworkers at Lehigh University, where PIV has been used to investigate a number of cavity-type geometries. Celik & Rockwell (2002) presented measurements of velocity, vorticity, and streamline patterns for shallow-cavity flows with and without the addition of a perforated surface covering the cavity. Their measurements provided a detailed view of the flow topology of the cavity at various phases. An example of their measurements showing the phase-averaged streamlines is presented in **Figure 3**. Similar measurements were presented by Celik et al. (2005), who investigated the effect of the thickness of the perforated plate. Continued work on perforated plates over cavities can be found in Ozalp et al. (2005) and Celik et al. (2007), who studied the influence of three-dimensional passive control geometries.

Haigermoser et al. (2008) recently obtained time-resolved measurements in a low–Mach number cavity with varied inlet boundary-layer thickness. This provided access to two-point velocity correlations as well as time-resolved forces acting on the cavity walls based on momentum. Noting the three-dimensional nature of the cavity flow, Haigermoser et al. (2009) later provided

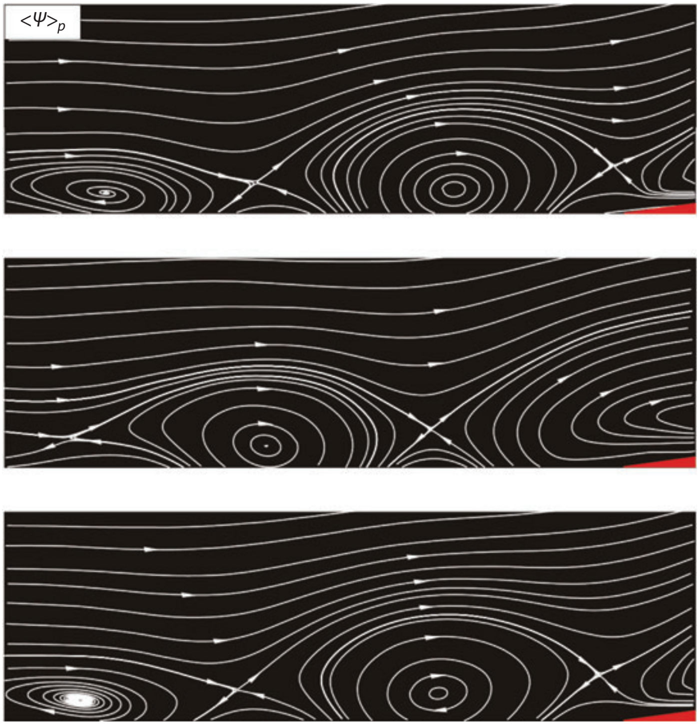



**Figure 3**

Phase-averaged streamlines showing flow topology for unstable flow past a cavity. The region of the images is defined by $0.3 < x < 1$, $0 < y < 0.3$, where the origin $(x, y) = (0, 0)$ is the separation point, and $L$ is the streamwise length of the cavity. Figure taken from Celik & Rockwell (2002).

tomographic PIV measurements (three components of velocity obtained in a volume of fluid). Analysis of the data obtained in these references in terms of acoustics was described by Haigermoser (2009). The momentum equation was used to obtain unsteady pressure from the velocity field, and this was combined with Curle's analogy to find the acoustic source terms. The main source of sound was found to be the downstream edge of the cavity at which the unsteady pressure was greatest owing to the oscillating vorticity field impinging on the edge.

Kang & Sung (2009) also provided a detailed study of a rectangular cavity in which proper orthogonal decomposition (POD) was used along with conditionally averaged two-point correlations to investigate the large-scale structures that result from the shear-layer instability. The structure of the shear-layer instability, including the number of vortex motions simultaneously present over the cavity, was described with varied inlet Reynolds number and varied cavity length. Qualitative interpretations of sound generation based on the instability mechanisms can be obtained from the discussion, although a direct assessment of the acoustic source terms was not provided.

## 2.2. Grazing Flow over a Helmholtz Resonator or Deep Cavity

A Helmholtz resonator is a device in which a volume of compressible fluid is enclosed by rigid boundaries with a single, small opening, as shown in **Figure 4**. The resonator is often modeled by a second-order mass-spring system analogy in which the fluid in the region of the orifice acts as

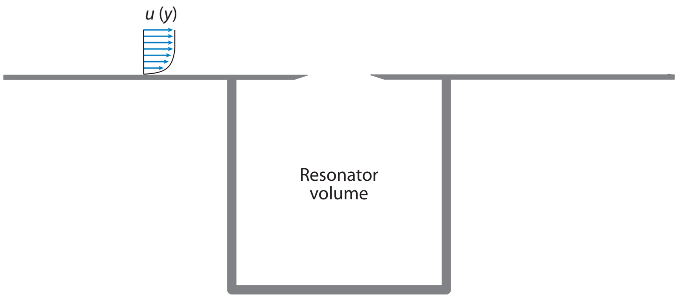



**Figure 4**

Schematic of the flow-excited Helmholtz resonator.



an effective mass, and the compressibility of the fluid in the volume acts as the stiffness. Acoustic radiation and viscous effects both can lead to damping. Unlike the purely hydrodynamic flow described in the previous section, the resonator has a natural frequency, $f_o$. Near this frequency, a small pressure disturbance can produce a large-magnitude velocity fluctuation at the orifice and thus a large pressure fluctuation within the resonator. In general this occurs at a low Mach number, and hence the wavelength of the resulting acoustic wave is large compared to the dimensions of the resonator.

The shear layer is linearly unstable to disturbances at discrete values of $fL/U$, where $f$ is the frequency of the oscillation, $L$ is the streamwise length of the orifice, and $U$ is the free-stream velocity (see, e.g., Howe 1998). Flow-excited resonance takes place when one of the hydrodynamic instability frequencies occurs at the same dimensional frequency as $f_o$.

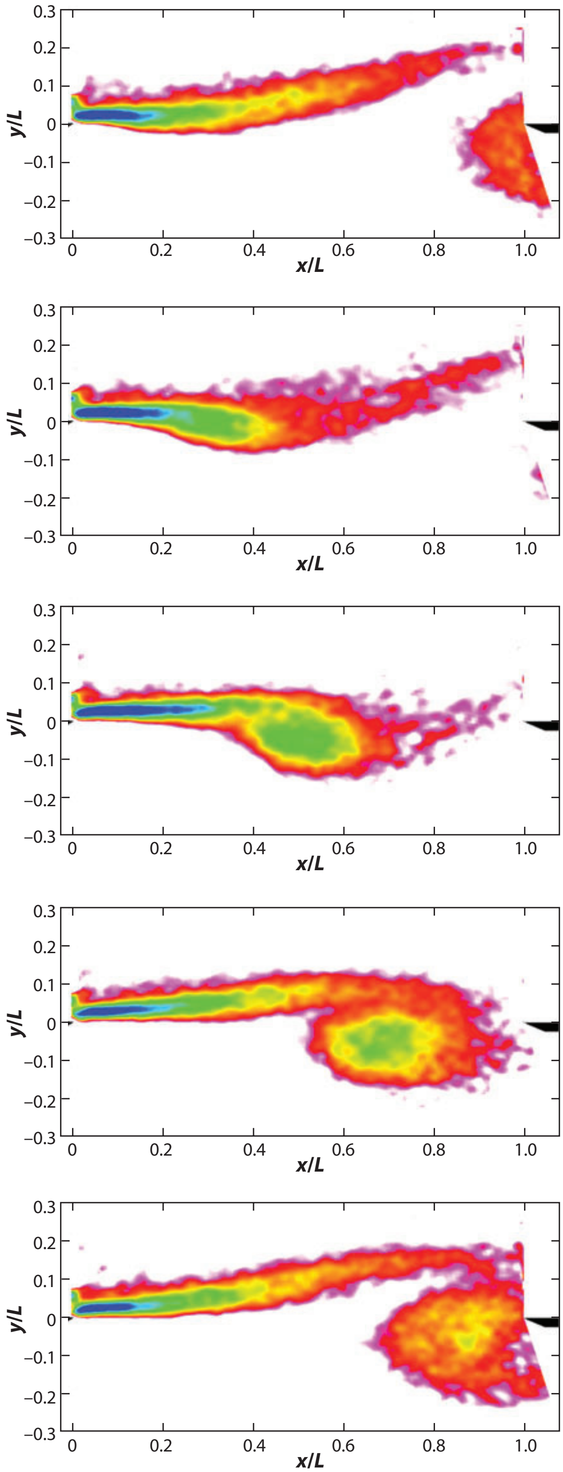
A recent study by Ma et al. (2009) described a flow-excited resonator using microphones and PIV. They also used a momentum-equation-based analysis to understand the relationship between the shear-layer instabilities and the resonator unsteady pressure. They found that the unsteady circulation (i.e., the area integral of spanwise vorticity over the orifice region) acted as a forcing function to the resonator. An example of the unsteady vorticity field obtained from phase-averaged PIV measurements is shown in **Figure 5**. These data show an interesting feature related to the evolution of the vorticity over the orifice. Specifically, many analytical models have been derived with the assumption that the shear layer either rolls up into compact, discrete vortex motions or, alternatively, acts as a continuous oscillating vortex sheet. The results shown in **Figure 5** suggest that the actual vorticity field can be interpreted to be in between these two analytical models. Ma et al. (2009) presented predictions of the resonator pressure based on the PIV results and found general agreement to within 2 dB over a wide range of flow speeds.

One of the applications of grazing flow over a Helmholtz resonator is the phenomenon known as window buffeting. This typically occurs in passenger vehicles when one window (or a sunroof) is lowered such that the cabin volume acts as the resonator. Kook (2009) and Slaboch et al. (2009) both provided PIV measurements in full-scale vehicles to describe the velocity/vorticity field that contributed to the cabin acoustic pressure.

The flow over a deep cavity, or side branch, forms a similar class of the cavity problem in which a feedback mechanism is present owing to the standing acoustic waves in the cavity. These organ-pipe-type modes occur at frequency values that are integer multiples of the fundamental, $c_o/L$, where $c_o$ is the speed of sound in the fluid, and $L$ is the depth of the side branch. The side branch can then result in flow-acoustic resonance when the shear-layer instability frequencies are near the side-branch natural frequencies. A detailed PIV study of a dual-side-branch flow was described by Oshkai et al. (2004, 2008) and later by Velikorodny et al. (2010). These papers used

**Figure 5**

Phase-averaged vorticity measurements in a flow-excited Helmholtz resonator. Figure taken from Ma et al. (2009).



detailed measurements of the velocity field along with Howe's acoustic analogy to derive estimates for the acoustic source magnitudes. The spatial distributions of acoustic source power showed a complex arrangement of sources and sinks of sound that depended greatly on the length of the cavity.

Yang et al. (2009) also provided microphone and PIV measurements in a side-branch resonator configuration. Their primary focus was the effect of the streamwise length of the opening. The length of the cavity was varied, and found to have a significant impact on the lock-on behavior of the cavity. Specifically, the hydrodynamic mode number (i.e., the number of vortex motions over the cavity) was sensitive to the velocity, cavity length, and the cavity depth. They found a minimum length required for the onset of a cavity tone. At larger cavity lengths, either one, two, or three vortex motions were observed over the cavity even with a turbulent approach boundary layer. The quantitative images show a relatively complex relationship between the shear-layer instability mechanisms and the geometry of the cavity. It is noted that Yang et al. focused their efforts on microphone measurements and PIV results, but did not consider the spatial distribution of acoustic sources.



## 2.3. Rossiter's Flow

The flow of a compressible fluid over a shallow cavity was originally studied by Rossiter (1964). This flow exhibits a shear-layer disturbance that grows over the cavity opening and impinges on the cavity trailing edge. This interaction results in a large pressure fluctuation that travels upstream as an acoustic wave, which leads to self-sustained oscillations. The resonance, or Rossiter mode, exists when the instability timescale of the shear layer is close to the acoustic travel time from one end of the cavity to the other. As a result, the acoustic wavelength and the orifice length are necessarily of the same order in magnitude, and the acoustic feedback travels in a direction parallel to the plane of the shear layer. The Rossiter mode is typically observed in cavities at Mach number values between 0.4 and 1.

Ukeiley & Murray (2005) acquired PIV measurements in a resonant cavity with two values of the cavity depth. The data were processed by stochastic estimation to use unsteady surface pressure to predict the time-dependent behavior of the velocity field. Two types of shear-layer instability were identified from the velocity/pressure measurements. One was found to be linked with the acoustic feedback and the other with the shear-layer structures propagating downstream. The distinction between these instability mechanisms was likely enhanced by the relatively thick boundary layer (9.4 mm) compared to the cavity depth (14.7 mm and 41.4 mm).

Hirahara et al. (2007) collected both Schlieren and PIV measurements in a high–Mach number cavity flow, providing additional descriptions of the unsteady vorticity field in the cavity. Numerous details were examined regarding the speed and trajectory of the vortex motions, streamline patterns, and mean flow quantities. The data were primarily used for a qualitative description of the flow physics in the cavity, and no direct assessment of the acoustic sources or instability mechanisms was provided.

PIV and unsteady wall pressure were also described by Murray et al. (2009) for five free-stream Mach numbers from 0.19 to 0.73. In this study the boundary-layer thickness was of the order 2 mm, compared to the cavity depth of 8 mm. The mean properties of the shear-layer development, growth, and trajectory were studied. The shear layer was found to exhibit a higher growth rate with increasing Reynolds number. Also, the recirculation zone inside the cavity was found to move downstream with increased Mach number. The focus of the analysis, however, was on the decomposition of the velocity measurements using snapshot POD. The relative weights of the dominant mode decreased with increasing free-stream Mach number and ranged from 15% to 20% of the total turbulent kinetic energy.

## 3. EXTERNAL FLOW

Flow over an aerodynamic object at all but the lowest Reynolds numbers leads to flow instabilities, turbulence, and hence radiated sound. PIV measurements have taken a leading role in the past several years in documenting the flow-field characteristics that lead to this mechanism of sound generation. This is often accomplished through an acoustic analogy such as Equation 4, in which the sources can be considered as unsteady surface pressures or Reynolds stress terms.

Jacob et al. (2005) presented PIV, radiated sound, and computational results for a rod-airfoil geometry. In this setup, a circular cylinder is placed in a uniform flow with a NACA 0012 airfoil located downstream in the cylinder wake. This problem is of interest in part because the unsteady velocity field in the cylinder wake exhibits large coherent motions that result from vortex shedding. These coherent motions impinge on the airfoil leading edge, resulting in relatively large unsteady forces and sound. An example of the flow near the leading-edge region is presented in **Figure 6**. The data show significant structure in the cylinder wake as it approaches the leading edge. The contours of normalized angular momentum (see Jacob et al. 2005) also show a significant interaction of the vortex structure with the airfoil, as noted by the large vortical region on the upper surface. A detailed analysis based on individual realizations, POD, and the two-point correlations of velocity has been used to describe detailed three-dimensional structures, secondary flows, vortex splitting, and stretching. These features of the flow were connected to the broadening of the spectral peak in the radiated sound. Jacob et al. also described Reynolds-averaged Navier-Stokes and large-eddy simulation calculations that were used along with PIV to understand and predict the radiated sound.

The influence of a cylinder wake on a NACA 0018 airfoil was also studied by Takagi et al. (2006). These authors also used a variety of tools, including PIV, radiated sound pressure, and

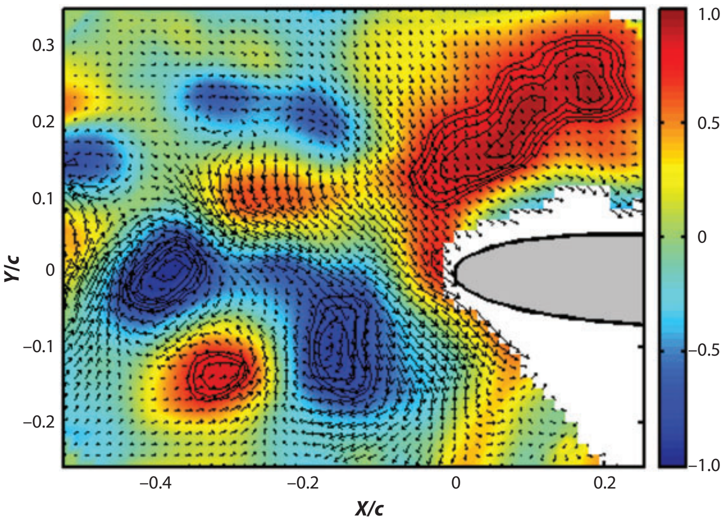



**Figure 6**

Instantaneous velocity vectors near the leading edge of an airfoil that was placed downstream of a circular cylinder. Color contours represent angular momentum. Figure taken from Jacob et al. (2005).

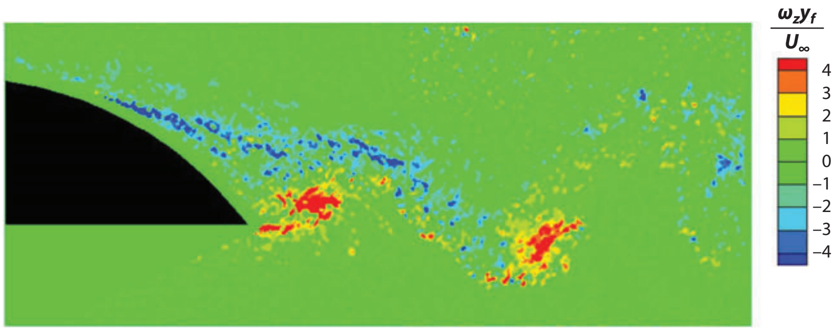



**Figure 7**

Instantaneous spanwise vorticity in the near wake of a blunt trailing edge. Figure taken from Shannon & Morris (2006).



liquid-crystal coatings, to visualize the laminar-turbulent transition. The transition and mean flow differences around the airfoil were studied in connection with the tonal sound generated by the interaction. Nakano et al. (2007) continued this work by using PIV to study the mean flow changes of the same airfoil at moderate Reynolds number at a varied angle of attack. At zero angle of attack the flow was nearly steady, and the acoustic pressure was negligibly different from the background noise. A spectral tone was observed in the radiated sound at moderate angles of attack (three to six degrees). The PIV data revealed that the tone was a result of a periodic instability that corresponded to the roll-up of the pressure side vorticity in the presence of the thicker suction side boundary layer. At higher angles of attack, the suction side was fully separated. As a result the pressure side vorticity was nearly steady in the vicinity of the trailing edge, and thus the acoustic tone was not present.

Shannon et al. (2006) and Shannon & Morris (2006) used PIV to investigate noise generated at the trailing edge of an airfoil. The model had a 45° beveled trailing edge that exhibited both broadband sound and vortex shedding. The trailing-edge sound was isolated using a pair of microphone arrays with a beam-forming algorithm that rejected other noise sources related to the airfoil or the wind tunnel (see Shannon & Morris 2008). The PIV data were processed to provide turbulent statistics that were phase averaged to the vortex shedding in the wake. An example of the instantaneous vorticity field is shown in **Figure 7**. Note that the vorticity field from this asymmetric geometry is very similar to that measured by Nakano et al. (2007) with a symmetric foil at a small angle of attack. The phase-averaged vorticity field is presented in **Figure 8**. The measurements show that the vorticity from the upper surface separates upstream of the sharp trailing edge and convects away from the airfoil. The vorticity in the lower surface periodically rolls up into a compact region and convects away from the edge. From these data it was hypothesized, and verified, that the broadband sound generated by the airfoil was amplitude modulated at the vortex-shedding frequency. Shannon & Morris (2006) provided additional measurements along with two-point correlations of the velocity field that further illustrate the amplitude-modulated nature of the sound sources.

Macumber et al. (2007) and Huyer et al. (2008) examined an airfoil with trailing-edge articulation. PIV was used to provide measurements of the unsteady velocity and vorticity fields in the near-wake region. The authors examined unsteady forces to identify sources of noise and found that the trailing-edge modifications altered the flow such that the estimated noise source terms were lowered by 5–8 dB.

Henning et al. (2008) measured the velocity field in the vicinity of a circular cylinder and an airfoil leading-edge slat simultaneously with radiated sound. Howe's vorticity-based acoustic analogy was used to describe the source terms and model the radiated sound field. Detailed

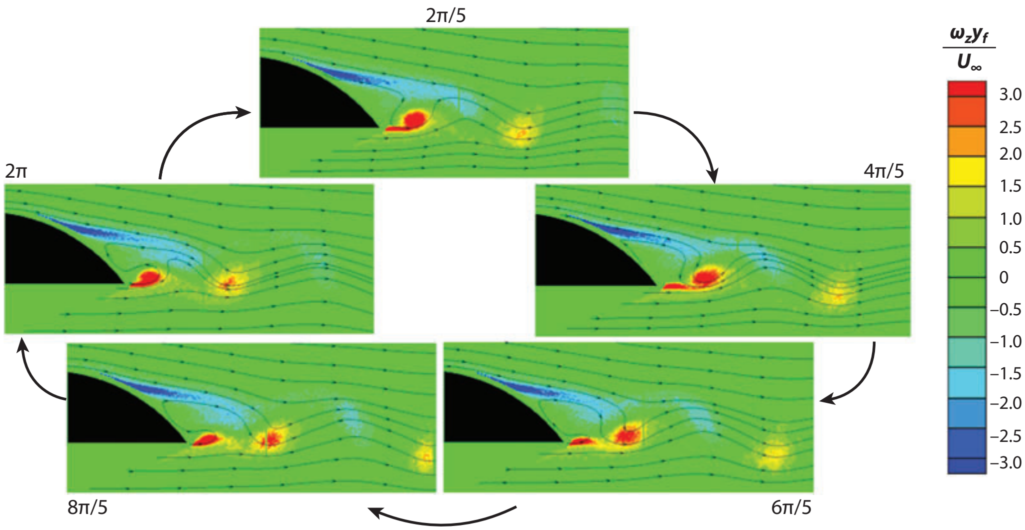



**Figure 8**

Phase-averaged vorticity in the near wake of a blunt trailing edge. Figure taken from Shannon et al. (2006).

calculations included POD modes and two-point correlation functions. In doing so, correlations were made between velocities, POD modal amplitudes, and far-field sound pressure. They found significant correlations between the sound, POD modes, and local flow velocities for the circular cylinder due to the large spanwise correlation length scale of the flow. The airfoil slat showed little or no correlation between the measured sound and the local velocity field. This was attributed to the complex and three-dimensional nature of this flow, which results in a very limited spanwise correlation length scale.

The flow around a circular cylinder results in unsteady lift and drag forces on the cylinder. A review that covers many aspects of this flow field is given by Williamson (1996). PIV measurements have been used to estimate the unsteady forces on cylinders by, for example, Unal et al. (1997). A number of related geometries have been considered in the context of unsteady forces on cylinders. Ozgoren & Rockwell (2007) and Rockwell (2008) investigated the unsteady flow over a short circular cylinder with free-surface wave effects. Sueki et al. (2010) used microphone measurements and PIV results to investigate how porous surface materials reduce the radiated sound. In all these examples, the PIV measurements were used to provide a qualitative description of the aeroacoustic source terms.

## 4. JET FLOW

The instabilities of jet flows are well studied, as is the sound created by jets at high speeds. The sound radiated by free jets is dominated by the volumetric distribution of $T_{ij}$ (see Equation 2). Thus an understanding of the noise sources necessarily requires both a spatial and temporal description of both the velocity and density fields. PIV measurements generally provide only a small fraction of the information required to describe the acoustic sources. The high speed and small scales of jet turbulence also limit how effective PIV can be for the study of jet noise. Seeding of the flow can also be particularly problematic and can lead to biased results (see Thurow et al. 2008). Nonetheless, PIV has been used to advance our understanding of jet flow physics and noise.

Schram & Riethmuller (2002) used a stroboscopic PIV technique in which the laser timing was synchronized with the jet instability frequency. The speed and Reynolds number were relatively low, such that the shear-layer instabilities were manifest as discrete vortex rings. The authors were able to investigate a complex vortex-pairing phenomenon in which the unsteady acceleration, threading, and merging of the vortex rings were documented. The intent of these measurements was to provide information regarding the vortex motion so that this mechanism of sound generation could be modeled and predicted, although no acoustic analysis was provided. It is noted, however, that Bridges & Hussain (1987) and others have shown that this is not likely to be a dominant source of noise in high–Reynolds number jet flows.

Bridges & Wernet (2003) assembled a system that was comprised of two PIV systems in order to obtain measurements in a heated jet facility. The use of two systems with controlled laser triggering allowed both space and time correlations of velocity to be computed. The paper was focused on the measurement technology and the validation of the data, but did not compute or describe the measurements in the context of the acoustic sources. Timmerman et al. (2009) obtained time-resolved measurements using large-scale gas-turbine engine exhaust. The PIV data were obtained with megahertz-range acquisition frequency to acquire temporal and spatial correlations of the velocity field along with near-field sound pressure. Timmerman et al. (2009) also focused mainly on the unique aspects of the measurement technology and have yet to use their data for the characterization of noise sources.

PIV and acoustic measurements of jet flows with chevrons were provided by Bridges & Brown (2004). Results from a baseline axisymmetric jet were compared with those from nine different chevron geometries. The mean streamwise component of velocity and vorticity were studied and used as a basis for the interpretation of the sound spectra and directivity patterns. The size and number of chevrons varied the penetration of the streamwise vortex motions, which yielded varied acoustic results. In general, the spectra showed lower sound levels at low frequency, and higher levels at high frequency. The effectiveness depended on the chevron count and penetration depth. Hot and isothermal jets were found to have similar trends. The mean velocity measurements were later used by Suzuki & Colonius (2006) for a linear stability analysis that was used in conjunction with a microphone array to detect amplitudes of instability waves.

A thorough study of jet flow physics and radiated noise is given by Alkislar et al. (2007) in a jet with $M = 0.9$. Stereo-PIV (planar, three-component) measurements were obtained at various downstream locations within the first few diameters of the jet. An acoustic array was used to provide sound source localization within the jet. Three different jet configurations were considered, including a baseline axisymmetric jet, the application of microjets to the baseline case, and a nine-node chevron geometry. An example of the acoustic results is shown in **Figure 9**. These data represent beam-forming maps of the acoustic sources based on the microphone array measurements. The microjet and chevron cases were generally found to decrease the noise at low frequency and increase the noise at the higher frequency, in agreement with the findings of Bridges & Brown (2004). At the higher frequencies shown, the microjet and chevron cases not only increase the magnitude of the sound, but also move the sources closer to the nozzle exit. Detailed turbulent statistics within the jet were examined in the context of how the jet produces noise. The mean velocity profiles, for example, are shown in **Figure 10**. The baseline case shows a primarily axisymmetric jet development (except for a small disturbance near the jet exit). The microjet case shows a distinctive perturbed shear-layer pattern near the jet exit that evolves to a wavy pattern downstream that persists far downstream of the jet exit. The chevron case shows a highly corrugated pattern at the jet exit. These large disturbances lead to a rapid growth of the shear layer and a nearly axisymmetric mean velocity field at a streamwise location of about three jet diameters. Based on these observations and other statistics, Alkislar et al. argued that the counter-rotating

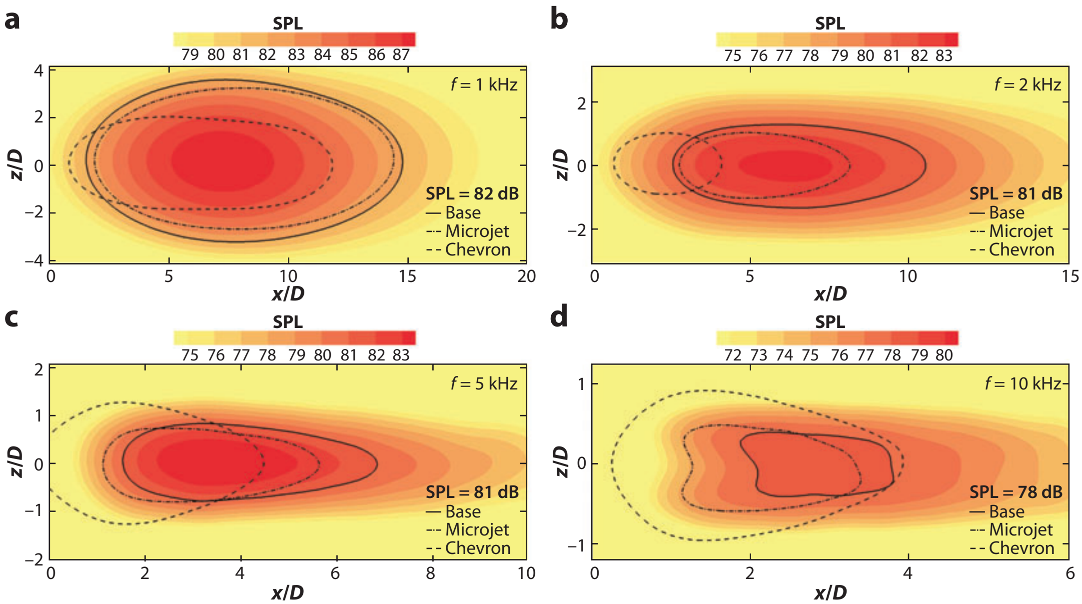



**Figure 9**

Spatial maps of noise production by an acoustic phased array at $fD/U$ values of (*a*) 0.25, (*b*) 0.5, (*c*) 1.25, and (*d*) 2.5. The contour lines correspond to a constant sound pressure level (SPL) of (*a*) 82 dB, (*b*,*c*) 81 dB, and (*d*) 78 dB. Figure taken from Alkislar et al. (2007).

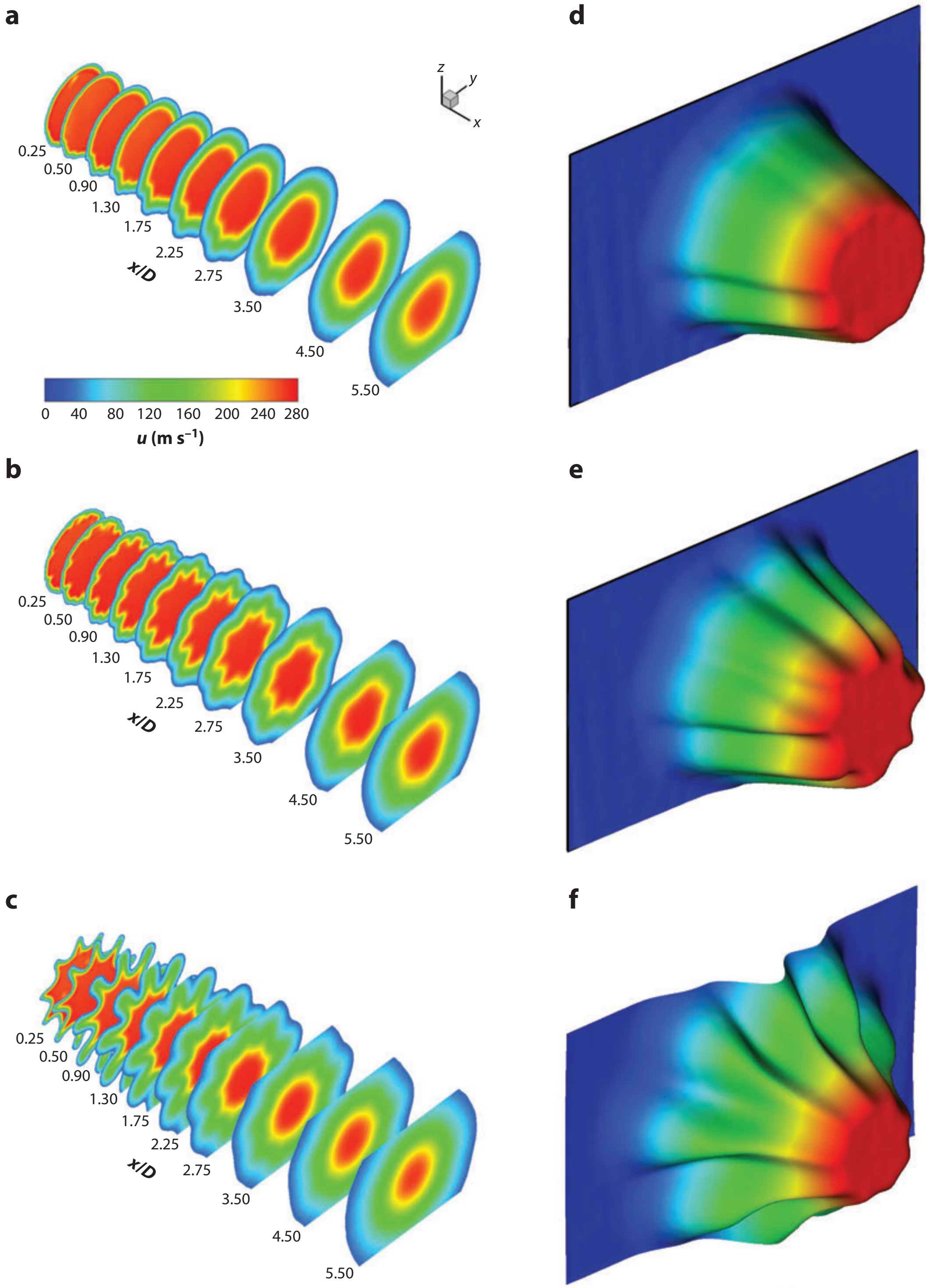
streamwise vortex pairs generated by the microjets were located at the high-speed side of the initial shear layer. The chevrons generated vorticies of greater strength on the low-speed side and thus decayed more quickly in the streamwise direction. The greater persistence and lower strength of the streamwise vorticity generated by microjets were attributed to the lowered high-frequency content of the radiated noise.

A study of active control in a high-speed subsonic jet using localized arc filament plasma actuators was given by Kearney-Fischer et al. (2009). The authors used PIV to observe the velocity statistics with various types of modal forcing. Specifically, the localized arc filament plasma actuators were arranged in a circumferential pattern around the nozzle near the exit such that the mode order of the forcing could be controlled. An example of the streamwise velocity obtained is shown in **Figure 11** with no forcing (baseline) and with aximuthal modes $m = 0$ (axisymmetric), $m = 1$, and $m = \pm 1$ (flapping). These data, as well as the associated turbulent statistics, were used to describe the effectiveness of the forcing on the evolution of the jet. Although acoustic measurements are not described in the study, this type of forcing is often motivated by noise reduction.

Fleury et al. (2008) used a dual-PIV system to obtain velocity statistics in a subsonic jet at $M = 0.6$ and $M = 0.9$. The authors presented the solution to the Lighthill equation in a form similar to Equation 6 to motivate the need for multipoint spatial and temporal correlation

**Figure 10**

Velocity maps at various streamwise locations, (*a*) baseline, (*b*) microjet, and (*c*) chevron, and the velocity profiles at $x/d = 2$ for (*d*) the baseline, (*e*) microjet, and (*f*) chevron. Figure taken from Alkislar et al. (2007).


a
b
c
d
e
f
0.25
0.50
0.90
1.30
1.75
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u (m s−1)

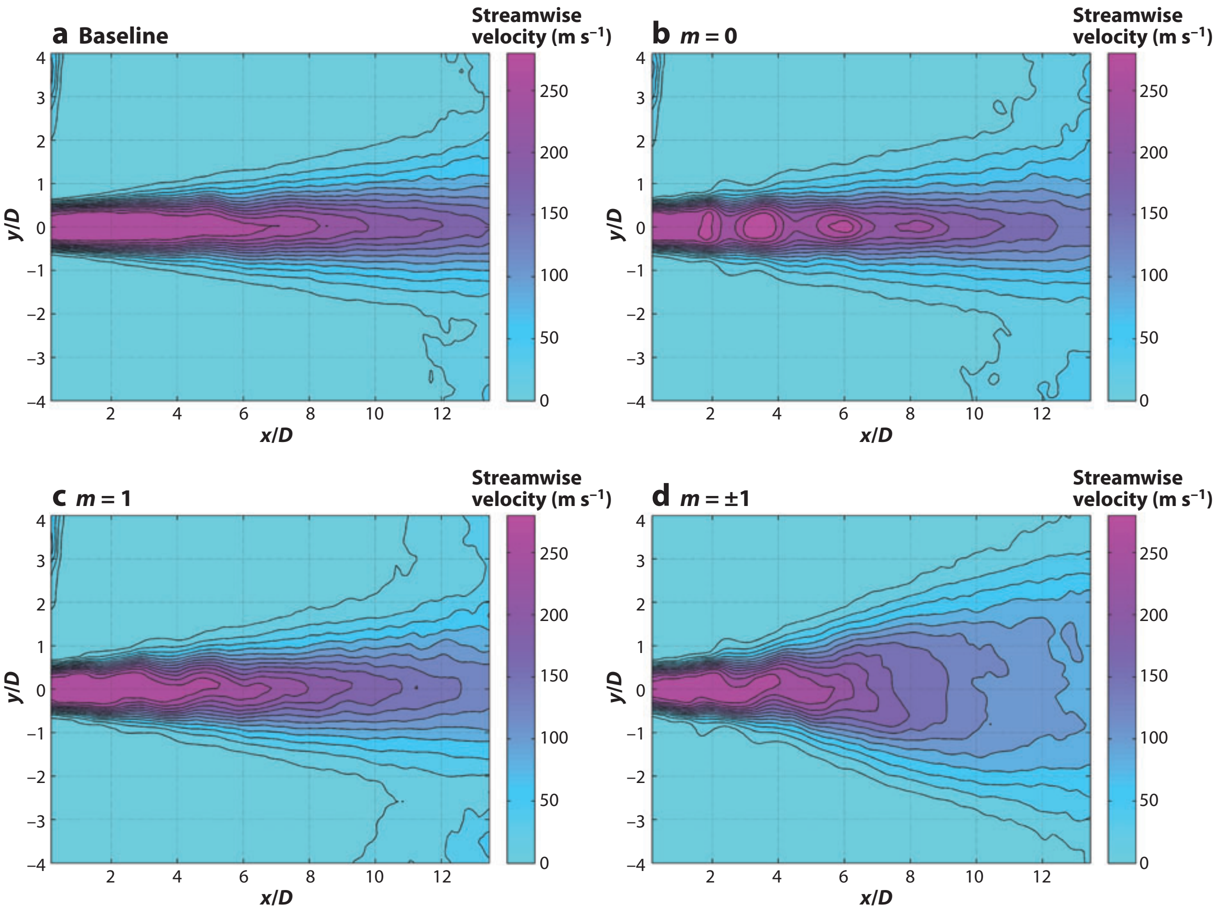


**Figure 11**

Conditionally averaged streamwise velocity for various forcing mode orders. The baseline condition represents no forcing. Figure taken from Kearney-Fischer et al. (2009).



functions. By employing two PIV systems together, the authors used a fixed time difference between acquisitions to obtain the temporal correlations. The shear layers of the jet were then analyzed over the first 11 jet diameters from the nozzle exit. The authors provided a very detailed discussion of the turbulent statistics, showing ratios of length scales and timescales as they evolve downstream to nearly isotropic turbulence values.

## SUMMARY POINTS

1. Shear-layer instabilities and turbulence are natural, unavoidable phenomena in many flows of interest and exhibit unsteady velocity, vorticity, and surface pressure. Current understanding of how sound is generated in aerodynamic flows is given by Lighthill's acoustic analogy or the various forms of the aeroacoustic equations that have been subsequently derived.

2. The availability of PIV systems to acquire high-resolution measurements has motivated its use in a large variety of flows to obtain measurements of velocity, vorticity, and fluid pressure. The three flows considered herein (the cavity, external aerodynamics, and jet flows) serve as three basic flows in which PIV has been used successfully in the study of sound production.
3. The sound sources can be represented statistically through a double convolution of a Green's function with the time-dependent two-point correlation of the Lighthill stress tensor. This type of information is never completely available in an experiment, and thus most PIV results are still used as a qualitative description of the acoustic sources.
4. The main goal of most PIV measurements as they relate to aeroacoustics is the measurement of the flow-field characteristics that lead to sound production. Direct measurements of turbulent flow characteristics are used to obtain amplitude, spatial distributions, and length scales of terms of the Lighthill stress tensor (the trailing-edge and jet-noise papers reviewed serve as good examples). However, the methods used to derive acoustic understanding from PIV measurements vary considerably depending on both the physical problem and the data available. Examples include inspection of mean flow properties to gain a qualitative understanding of the noise sources, and the use of two-point correlation measurements for more direct acoustic modeling. Other methods include phase averaging of the velocity data, POD analysis, and the calculation of dipole force estimates. Because a complete description of a flow is never experimentally available, a combination of these methods is most often used to gain a qualitative understanding of noise generation. As a result it is often difficult to compare the results of various studies even when the same flow field is being considered.
5. New advanced PIV methods may have a substantial impact on the study of acoustic sources in the near future. Two examples are tomographic PIV and time-resolved PIV. Continuing improvements in these technologies will provide more complete information about sound production.



## DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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